University of Mumbai
Examination May 2022

Examinations Commencing from 17th May 2022

Program: **Information Technology**

Curriculum Scheme: Rev 2019

Examination: SE Semester IV

Course Code: ITC404 and Course Name: AUTOMATA THEORY

Time: 2 : 30 hours Marks: 80

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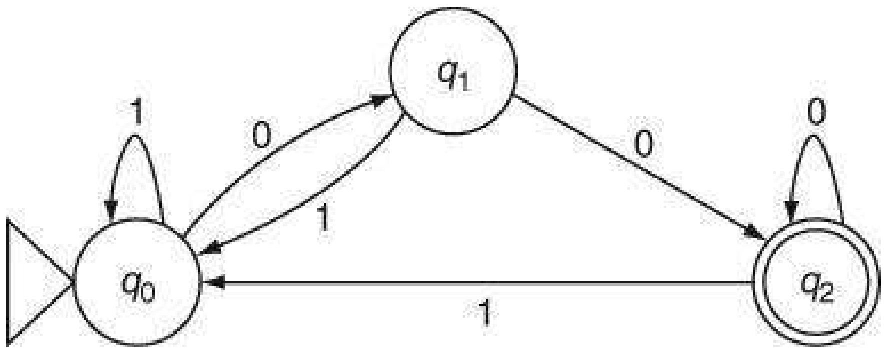
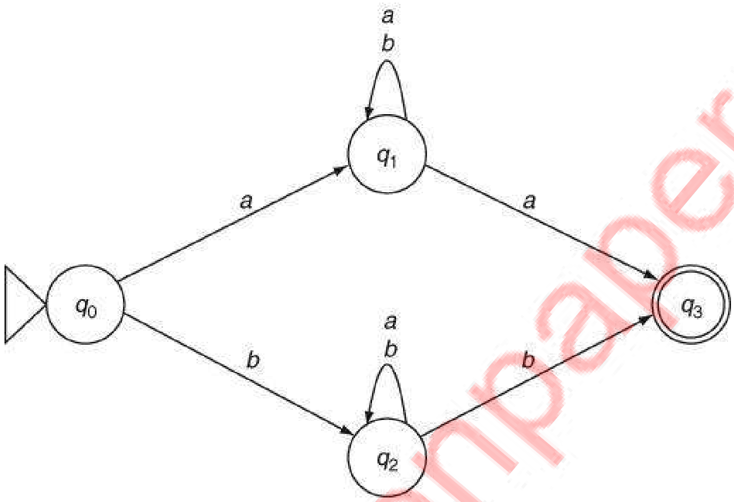
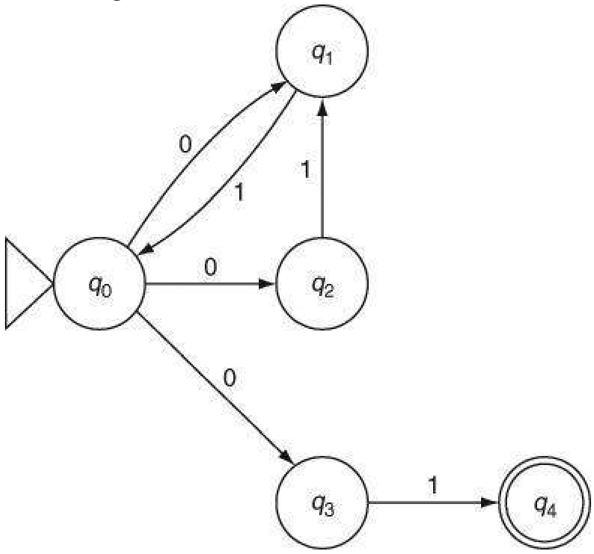
Q1.	**Choose the correct option for following questions. All the Questions are compulsory and carry equal marks 20M**
1.	Transition function of any automata defines
Option A:	Σ * Q –> Σ
Option B:	Q * Σ –> Σ
Option C:	Σ * Σ –> Q
Option D:	Q * Σ –> Q
2.	What is the correct form of production in Chomsky's Normal Form?
Option A:	$S \rightarrow aS$
Option B:	$S \rightarrow AB$
Option C:	$S \rightarrow Sa$
Option D:	$S \rightarrow A$
3.	Which of the following is a regular expression for binary strings with no consecutive 1's?
Option A:	(01+10)*
Option B:	(1+λ) (01+0)*
Option C:	(0+1)*(0+ λ)
Option D:	(10+0)*(1+ λ)*

4.	
Option A:	Given DFA is for binary numbers divisible 2
Option B:	Given DFA is for binary numbers divisible 3
Option C:	Given DFA is for binary numbers divisible 4
Option D:	Given DFA is for every 0 followed by 1
5.	
Option A:	Given DFA is for strings with the same first and last symbol
Option B:	Given NFA is for strings with the same first and last symbol
Option C:	Given NFA is for strings for searching the keyword “aba” or “bab”
Option D:	Given NFA is for strings with any combination of a’s and b’s
6.	**S→ 1S / 01 S** **S→ 0A** **A→ 0B** **B→ 1B / 10B / λ**
Option A:	The regular expression for above grammar is **(1 + 01)*00(λ + 0)***
Option B:	The regular expression for above grammar is **(1 + 01)*00(1 + 10)***
Option C:	The regular expression for above grammar is **(1 + 01)*)000(1 + 10)***
Option D:	The regular expression for above grammar is **(0 + 01)*0(1 + 01)***

7.	The grammar for the language where a's followed by twice as many *b*'s, i.e, $a^n b^{2n}$ Where n >=1 .
Option A:	S → aSbb | b
Option B:	S → aSbb
Option C:	S → aSbb | λ
Option D:	S → aSbb | ab
8.	What is the language of Finite Automata ?
Option A:	Recursive sensitive Language
Option B:	Regular Language
Option C:	Context Sensitive Languages
Option D:	Context free Language
9.	The ……………………is a programmable machine that can compute anything that is computable
Option A:	Deterministic Finite Automata
Option B:	Non Deterministic Finite Automata
Option C:	Universal Turing Machine
Option D:	Push down Automata
10.	Which of the following relates to the Chomsky hierarchy?
Option A:	Regular<CFL<CSL<Unrestricted
Option B:	CFL<CSL<Unrestricted<Regular
Option C:	CSL<Unrestricted<CF<Regular
Option D:	CSL<Unrestricted< Regular<CF

<table>
<tr><th>Q2.</th><th>Solve any Four questions out of Six. 5 marks each</th></tr>
<tr><td>A</td><td>Convert given NFA to DFA

</td></tr>
<tr><td>B</td><td>Construct only a Mealy machine for the following:
For input from, ∑*, where ∑= (0,1), if the input ends in ‘101’, the output should be ‘x’; if the input ends in ‘110’, output should be ‘y’ otherwise output should be ‘z’. (transition table and diagram both are expected)</td></tr>
<tr><td>C</td><td>Give Regular Expressions for
i) For all strings over 0,1 that starts with 10 and ends with 01
ii) For all strings over a,b which contains exactly 3 occurrence of ‘ b’ over Σ={a,b}</td></tr>
<tr><td>D</td><td>Consider the following CFG: G = { (S, A), (a, b), P, S}, where P consists of :
S→aAS | a
A→SbA | SS | ba
Derive the string ‘aabbaa’ using leftmost derivation and rightmost derivation.</td></tr>
<tr><td>E</td><td>Compare and Contrast between FA, PDA and TM</td></tr>
<tr><td>F</td><td>what is Ambiguous Grammar, find if the following grammar is ambiguous or not by generating $(x + 2.0) * y / (z – 6.0)$

S → S + S
S → S * S
S → S – S
S → S / S
S → (S)
S → variable | constant</td></tr>
</table>

Q3.	**Solve any Two Questions out of Three 10 marks each**
A	What are steps for converting CFG to CNF ? Convert the given grammar G to CNF. G: **$S \rightarrow a \mid aA \mid B \mid C$** **$A \rightarrow aB \mid \varepsilon$** **$B \rightarrow Aa$** **$C \rightarrow aCD \mid a$** **$D \rightarrow ddd$**
B	Give a formal definition of Turing Machine (TM). Design a TM that performs the addition of two unary numbers. (transition table and diagram both are expected)
C	Design PDA for for odd length palindrome, **let $\sum = \{0,1\}$,** **$L = \{ W X W^R \}$**
Q.4	**Solve any Two Questions out of Three 10 marks each**
A	Explain Chomsky's Hierarchy with neat diagram
B	Construct DFA for given regular expression (a+b)* aba (a+b)*
C	Construct NFA with Ɛ moves for "Zero or more number of 0's followed by zero or more number of 1's followed by zero or more number of 2's . convert this to DFA.